

AI/ML Task 2: Model Comparison Report

Feature Engineering, Model Optimization & Performance Comparison

Objective

This task improves the Task 1 house price prediction workflow by adding preprocessing, feature engineering, feature scaling, multiple regression models, and performance comparison.

The goal was to compare models using measurable metrics and select the best-performing model using actual results from the executed notebook.

Dataset and Preprocessing

Dataset source: Uploaded CSV file named housing.csv.

Rows: 20,640. Columns after preprocessing: 13. Features after engineering: 14.

Target variable: HousePrice, representing median house value in 100,000 dollar units.

The notebook checked missing values, data types, descriptive statistics, feature distributions, correlations, and location-based price patterns.

Feature engineering added RoomsPerPerson and BedroomRatio. StandardScaler was used inside pipelines for Linear Regression and Ridge Regression to keep feature ranges comparable.

Models Compared

- Linear Regression: baseline regression model.
- Ridge Regression: regularized linear model to reduce overfitting risk.
- Decision Tree: nonlinear model with max_depth=5.
- Random Forest: ensemble model with 100 estimators and max_depth=12.

Results and Model Selection

Model Performance Comparison

Model	MAE	RMSE	R2 Score
Random Forest	0.339168	0.515623	0.797111
Linear Regression	0.482436	0.673856	0.653481
Ridge Regression	0.482428	0.673860	0.653477
Decision Tree	0.496911	0.712963	0.612094

Best Model

The best-performing model was Random Forest. It had the lowest RMSE, lowest MAE, and highest R2 Score among the tested models.

Best model metrics: MAE = 0.339168, RMSE = 0.515623, R2 Score = 0.797111.

The lower RMSE shows that Random Forest made smaller prediction errors compared with the other models. The R2 Score of 0.797111 means it explained about 79.7 percent of the variation in the test data.

Validation and Interpretation

The actual vs predicted plot showed that Random Forest predictions followed the correct price trend more closely than a basic linear baseline. Residual plots were used to inspect error patterns.

Cross-validation R2 scores for the selected model were 0.427486, 0.628305, 0.722829, 0.486803, and 0.737046. The mean CV R2 was 0.600494 with a standard deviation of 0.124293.

Conclusion

Task 2 demonstrates that real ML workflows should not stop at one model. Feature engineering, scaling, multiple algorithms, and metric-based comparison help justify the final model choice.

Random Forest was selected because it performed best on the test set. The model was saved using joblib as task2_best_house_price_model.pkl for future reuse.